

Algebra 1
Unit 2
Lesson 3 Practice

Name _____
Date _____
Period _____

1. Pam was simplifying the expression and made an error. Her steps are shown. Find and fix the mistake.

$$\left(\frac{4x^3y^3 \cdot 2xy}{3xy^2}\right)^2$$

Step 1: $\left(\frac{8x^4y^4}{3xy^2}\right)^2$

Step 2: $\frac{64x^6y^6}{9x^2y^4}$

Step 3: $\frac{64x^4y^2}{9}$

2. Write a note to Pam explaining her error.

For questions 3 through 5, find an equivalent expression using the laws of exponents.

3. $\frac{14x^6y^6}{12x^{10}y^3}$

4. $\frac{4x^6y^6}{6x^2} \cdot \frac{3x}{4y}$

5. $\left(\frac{4x^2y^3}{5x^0y^5}\right)^2$

6. Prove $2^{(2x+4)} = 16 \cdot 2^{(2x)}$ by using the laws of exponents.

7. Prove $4^{(x-2)} = \frac{4^x}{16}$ by using the laws of exponents.

8. Write an equivalent expression for $(-3.1)^{(4x)} \cdot (-3.1)^{(2x+7)}$

9. Use the law of exponents to write an equivalent expression for $5^{(7-3x)}$.

10. Use the law of exponents to write an equivalent expression for $(2.8)^{(6x+8)}$.